

A Comparative Demographic and Economic Study: Bihar vs. Kerala

Demography and Economic Statistics Project

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1 Introduction

The National Family Health Survey (NFHS) provides one of the most comprehensive data set to monitor health, demographic, and socioeconomic trends in India. This project is based on the NFHS unit-level data to conduct a detailed comparative analysis between two states representing the two extremes of India's development and demographic spectrum: **Bihar**, a state characterized by a slow demographic transition and **Kerala**, a pioneer of demographic progress often cited as a model for successful social development.

The core objective of this project is to dive deeply into the factors driving the demographic difference by meticulously examining how different socioeconomic factors influence various aspects of living standards in two radically different policy environments. We analyze a subset of NFHS data, including core demographic variables and socioeconomic indicators, using sophisticated statistical methods to draw robust and policy-relevant conclusions.

The research is divided into five distinct modules, each addressing a critical aspect of the Bihar-Kerala contrast:

2 Research Questions

1. Among women who already have one child, what factors predict progressing to have a second?
2. How does various socio-economic factors influence the total children ever born by a woman?
3. How does various social determinants as wealth and education affect the possibility of improved sanitation in household?
4. How are wealth quintile and education level of women in household interrelated?
5. How does household wealth affect the odds of early childbearing?

3 Research Objectives

The broad objective of this project is to conduct a comparative quantitative analysis of the socio-economic determinants influencing key demographic and health outcomes in Bihar and Kerala, using unit-level data from the National Family Health Survey (NFHS-5).

The specific objectives, corresponding to each research question, are:

1. To identify the socio-economic factors (including **wealth**, **education**, and **age**) that predict **parity progression** from a first to a second child, a second to a third child or a third to a fourth child and to test whether the influence of these factors varies significantly between Bihar and Kerala.
2. To investigate the influence of key socio-economic factors (primarily **wealth**, **education**, and **age**) on the **total children ever born** (TCEB) by a woman, and to compare the statistical significance and effect size of these determinants in Bihar versus Kerala.
3. To assess the impact of social determinants, specifically **household wealth** and **education**, on a household's access to **improved sanitation**, and to formally test if the effects of these predictors are statistically different between the two states.
4. To analyze the interrelationship between **household wealth quintile** and a woman's **educational attainment**, with a specific focus on using a modified Poisson model to correctly estimate effects in the presence of Kerala's **ceiling effect** in education.
5. To model and quantify the effect of **household wealth** on the odds of **early childbearing** (among women aged 15-21) in Bihar and Kerala, and to test if this effect differs significantly between the two states.

4 Research Hypothesis

Based on the established demographic literature concerning the divergent development paths of Bihar and Kerala, we formulate the following hypotheses, which correspond to our five primary research questions:

1. **Parity Progression:** We hypothesize that the influence of wealth and education on parity progression is not uniform, and that there is a significant statistical interaction between socio-economic status and state (Bihar vs. Kerala).
2. **On Total Children Ever Born:** We hypothesize that the socio-economic gradient for total children ever born will be steep and significant in Bihar, whereas in Kerala, this gradient will be flat and non-significant due to a **saturation effect** at replacement-level fertility.
3. **On Improved Sanitation:** We hypothesize that access to improved sanitation will be strongly predicted by household wealth and education in Bihar, but these factors will be largely non-significant in Kerala due to a 'ceiling effect' of near-universal access.
4. **On the Wealth-Education Link:** We hypothesize a strong, positive association between household wealth and achieving secondary education in Bihar, but we expect this link to be significantly weaker in Kerala, where universal access has decoupled higher education from wealth.
5. **On Early Childbearing:** We hypothesize that higher household wealth and education are significant protective factors against early childbearing, and we expect this protective effect to be substantially stronger in Bihar than in Kerala.

5 Data Sources

All the estimation, fitting of models and inference was done based on the microlevel Women's IR dataset for Bihar and Kerala collected under **NFHS-5** regime.

6 Factors Influencing the Transition to a Second Birth

6.1 Methodology

Under this section, we aim to investigate the factors predicting the progression to a second child among women who already have at least one. For this purpose, we restrict our dataset to respondents who have borne at least one children. The main factors of interest here whose influence we

will be testing are **household wealth index**, **education**, and **place of residence**, all of which are categorical variables and a continuous variable: **the respondent’s current age**.

To account for the complex survey design, survey design objects were created for Bihar (State 10), and Kerala (State 32), and the combined sample, specifying sampling weights. Weighted logistic regression models with a quasi-binomial family (to handle over-dispersion) were fitted separately for Bihar and Kerala to estimate adjusted odds ratios (AORs) for wealth, education, residence, and age. Subsequently, a combined model was fitted to the pooled data. This model included interaction terms for both wealth-by-state and education-by-state. This was done to formally test if the effects of wealth and education varied significantly between the two states. We used a formal Wald test to assess the statistical significance of these interaction terms. Finally, predicted probabilities from the interaction model were generated and visualized to illustrate any statewise differences.

6.2 Salient Findings and Discussion

Analyzing the parity progression from one to two children directly measures the strength of the “**two-child norm**”. It is the single most important lever for maintaining a replacement-level TFR (around 2.1). If this progression rate drops significantly (e.g., below 80-90%), it signals a societal shift to “**one-and-done**” families, leading to rapid, below-replacement population decline.

Similarly, a sharp societal drop in the progression from two to three children is the clearest indicator of a successful **demographic transition** from high to low fertility. This specific “**stopping behaviour**” is what moves a population’s TFR from above 3.0 down toward replacement. This progression is also a representative measure of son preference (a family with two daughters is far more likely to “**progress**” than one with two sons).

Finally, the progression rate from three to four children should be near zero in low-fertility populations. A high rate is a direct signal of an **unmet need for family planning** and contraception.

This behaviour is predominantly confined to the most vulnerable households, serving as a stark indicator of **deep-rooted poverty**, low female educational attainment, and inadequate access to healthcare.

Our investigation into the progression from a first to a second child reveals striking differences between Bihar and Kerala. A crude analysis of the weighted prevalence of having two or more children (given at least one) provides an initial look at this divergence. As shown in Table 1, the prevalence of progression to 2 children when having borne at least one in Bihar is extremely high, starting at 87.1% for the poorest quintile and showing a modest monotonic decline to 74.6% for the richest. In contrast, Kerala’s crude prevalence is lower and follows a non-monotonic pattern, hovering between 73% and 79% with no clear trend. The standard error for Kerala’s “Poorest” quintile is notably large (0.054) compared to Bihar’s (0.004), indicating significant data sparsity for this subgroup in Kerala, which was also confirmed by initial cross-tabulations.

Table 1: *Weighted Prevalence of Parity Progression (1+ to 2+ Children) by Wealth Quintile*

Wealth Quintile	Bihar (State 10) Prevalence (%)	Bihar (SE)	Kerala (State 32) Prevalence (%)	Kerala (SE)
1 (Poorest)	87.1%	0.0035	78.9%	0.0543
2 (Poorer)	86.9%	0.0044	75.2%	0.0242
3 (Middle)	85.5%	0.0059	75.9%	0.0130
4 (Richer)	82.7%	0.0090	77.3%	0.0083
5 (Richest)	74.6%	0.0195	73.4%	0.0094

To understand these patterns, we fitted separate weighted logistic regression models for each state, controlling for wealth, education, residence, and age. The Adjusted Odds Ratios (AORs) in Table 2 reveal that the drivers of parity progression are fundamentally different in the two states.

Upon analyzing the above data, we develop a rough intuitive idea that in Bihar, **wealth** becomes a significant protective factor at the highest quintiles. Compared to the poorest, women in the “Richer” quintile have 20% lower odds (AOR=0.797, $p < 0.05$) and women in the “Richest” quintile have 66% lower odds (AOR=0.342, $p < 0.001$) of having a second child. **Education** is an even

Table 2: Adjusted Odds Ratios (AORs) for Parity Progression (from Models 1 & 2)

Factor	AOR for Bihar	AOR for Kerala
wealth2	1.096	0.560
wealth3	0.985	0.655
wealth4	0.797	0.864
wealth5	0.342	0.830
education1	0.974	1.547
education3	0.596	1.581
education4	0.326	1.644
education5	0.314	0.604
residence2	1.148	1.006
age (continuous)	1.241	1.108

stronger, highly significant predictor; women with “complete secondary” (code 3) or higher (codes 4 & 5) education have 40-69% lower odds of progression. **Age** is a powerful positive predictor (AOR=1.24), indicating that for each additional year of age, a woman’s odds of having a second child increase by 24%. However, in Kerala, the results are at least apparently different. Neither **wealth** nor **education** appears to be significant in determining the odds of progressing to a second child.

Given these stark differences, we want to formally test whether the effects of wealth and education varies across states. To this end, we would be using a combined interaction model on data from both Bihar and Kerala.

The above table 3 tells us that wealth, education and age are the important factors that determine the progression to a second child. Further, we also note that the interaction terms of education-state and wealth-state, play a significant impact here. In order to test this statistically, we have performed a Wald test. For the test related to education-state interaction effect, we get the F-statistic as 11.95371 (on 4 and 2277 df) corresponding to a p-value of 1.2951e-09. However, for the test related to wealth-state interaction effect, we get the F-statistic as 12.69533 (on 4 and 2277 df) corresponding to a p-value of 3.2091e-10. Both the p -values are much smaller than 0.1 and hence we reject both the null hypothesis. Now, we can say statistically that the wealth-state and the education-state interactions are significant statistically and vary between the two states.

Table 3: *Combined Regression Analysis on data from both Bihar and Kerala*

Variable	Estimate	p-value
(Intercept)	-2.674868	< 2e-16
wealth2	0.128643	0.015682
wealth3	0.068574	0.281688
wealth4	-0.087550	0.297855
wealth5	-0.783763	2.76e-09
state32	-2.088687	0.000134
education1	-0.104363	0.141725
education3	-0.661397	< 2e-16
education4	-1.350428	9.81e-10
education5	-1.283975	< 2e-16
residence2	0.071897	0.239768
age	0.165773	< 2e-16
wealth2:state32	-1.028388	0.009352
wealth3:state32	-0.803751	0.036101
wealth4:state32	-0.327119	0.381161
wealth5:state32	0.245155	0.542178
state32:education1	0.571895	0.229038
state32:education3	1.581970	0.000100
state32:education4	2.552474	6.34e-07
state32:education5	1.334872	0.001503

Figure 1 provides a clear visualization of this significant interaction. The predicted probability of parity progression in Bihar (red line) is very high (around 85-88%) and largely flat, with a significant drop only for the richest quintile. The confidence intervals are extremely tight, reflecting the large sample size.

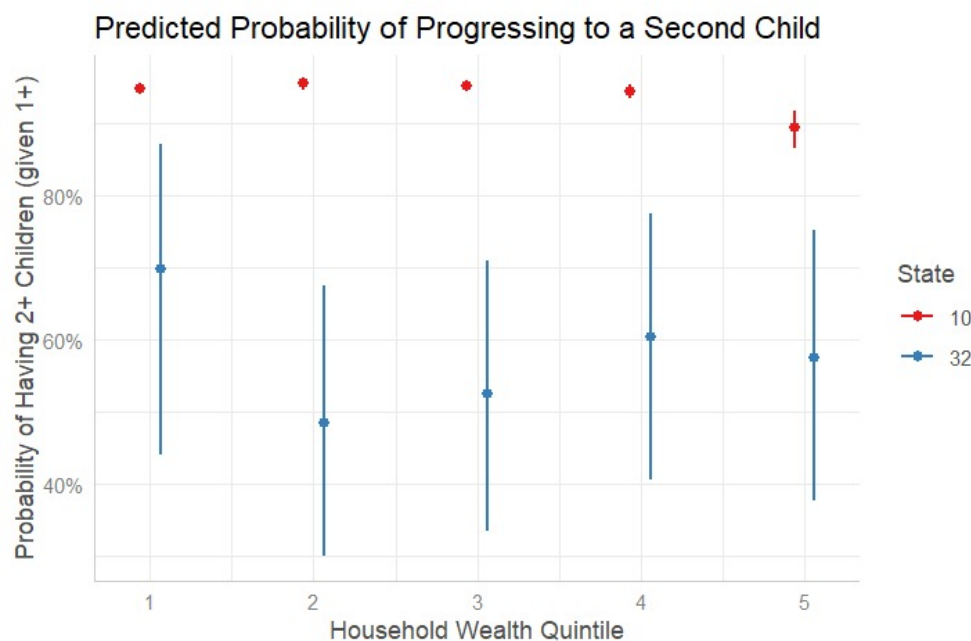


Figure 1: Predicted Probability of Progressing to a Second Child by Wealth and State

Conversely, the probability in Kerala (blue line) is not only lower overall but shows a different pattern, with a steep visual drop from the poorest to the poorer quintiles before flattening out. The confidence intervals for Kerala are substantially wider, especially for the poorest quintile, reflecting the data sparsity noted earlier. Nonetheless, the formal Wald test confirms that these two patterns are statistically distinct.

We can further extend our analysis in a similar fashion to check for **parity progression** from 2 to 3 children and from 3 to 4 children as well. In stead of presenting the entire analysis for **AORs** and interaction effects, we present the confidence interval plots up here that helps us to visualize the situation.

As seen in Figure 2, the two states show a complete divergence. In **Kerala (State 32)**, the probability of having a third child is extremely low (below 20%) and, crucially, **is flat across all wealth quintiles**. This reinforces the finding that the two-child norm is absolute and is not stratified by wealth. In **Bihar (State 10)**, a strong socio-economic gradient is evident. The probability of having a third child is very high for the poorest quintile (over 80%) and shows a

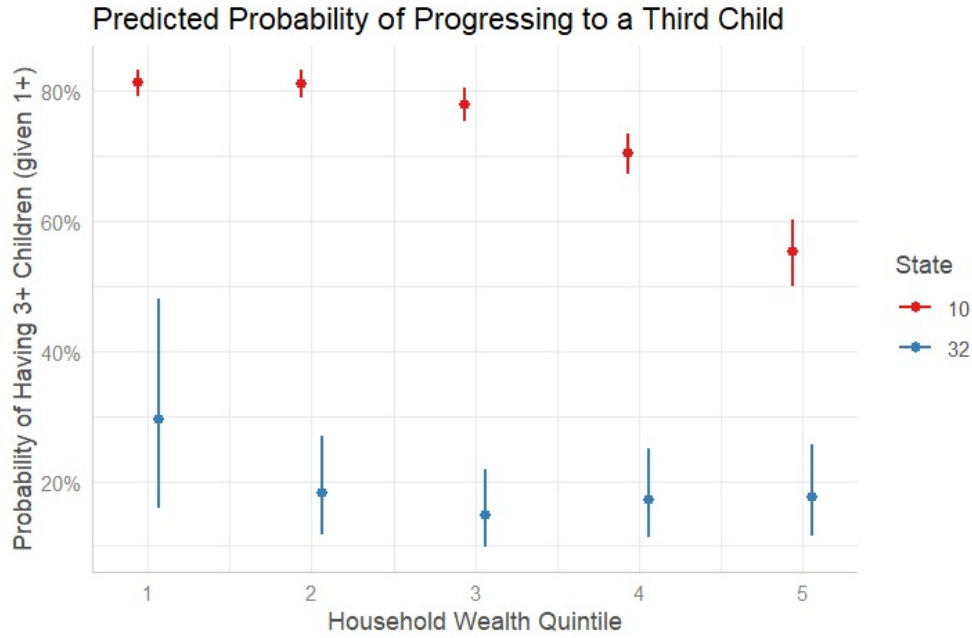


Figure 2: Predicted Probability of Progressing to a Third Child (Given 1+)

steep, monotonic decline as wealth increases, dropping to 55% for the richest.

Similarly, if we look at the plot for progression to a fourth child (Figure 3), we establish our point even more starkly. In **Kerala**, the probability is very low (less than 10%) for all groups. In **Bihar**, the wealth gradient is even steeper than for the third child. The probability drops from nearly 80% for the poorest quintile to just 25% for the richest. However, it is of great importance to note that unlike in the previous cases, the interaction effect of education and state in predicting progression to a fourth child is statistically negligible ($p\text{-value} \approx 0.38$), that is education serves almost similarly as a strong protective measure against progression to a fourth child in both the states.

6.3 Conclusion

Our analysis confirms that the socio-economic drivers of parity progression are fundamentally different in Bihar and Kerala. This visual intuition was validated by our combined regression models, where formal **Wald tests** for interaction terms were highly significant across multiple parity steps.

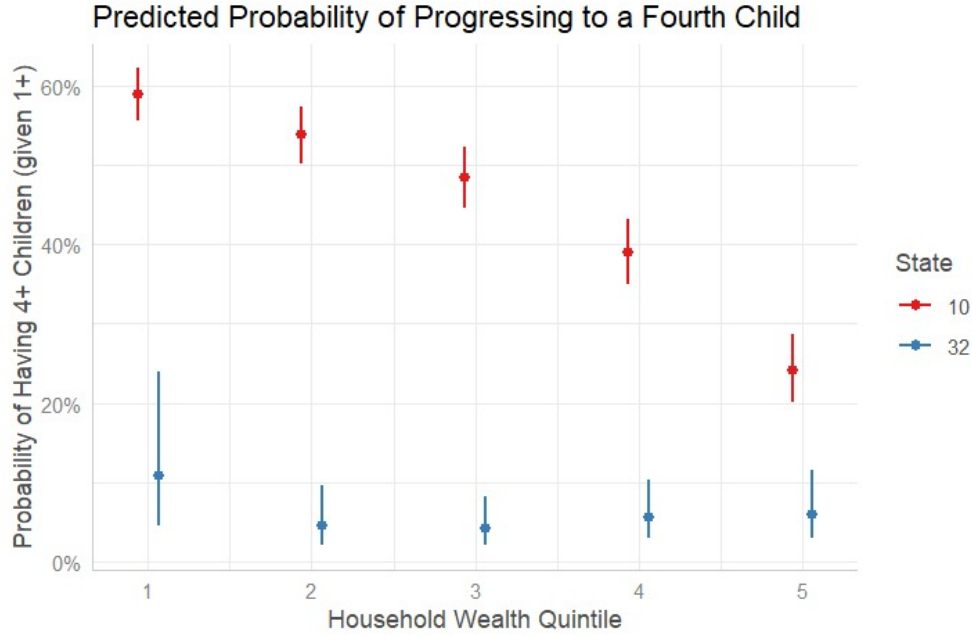


Figure 3: Predicted Probability of Progressing to a Fourth Child (Given 1+)

For the progression from **one to two children**, both the **wealth-by-state** ($p < 3.21 \times 10^{-10}$) and **education-by-state** ($p < 1.30 \times 10^{-9}$) interactions were highly significant. This divergence is reinforced for the **two-to-three child** progression, where both the **wealth-by-state** ($p < 4.79 \times 10^{-12}$) and **education-by-state** ($p < 4.97 \times 10^{-5}$) interactions also remain statistically significant.

This paints a stark portrait of two demographic regimes. Bihar remains clearly mid-transition, where fertility behaviour is highly stratified by class; higher wealth and education are strong protective factors that significantly *reduce* the odds of having more children. Conversely, Kerala exemplifies a **saturated** state where the demographic transition is complete. Here, the two-child norm is so universal that these socio-economic factors are no longer significant, suggesting that fertility decisions are decoupled from a household's wealth or education.

Tellingly, for the progression from **three to four children**, the **wealth-by-state** interaction remains a powerful differentiator ($p < 1.13 \times 10^{-8}$), while the **education-by-state** interaction is no longer significant ($p = 0.390$). This suggests that as number of children rise, education acts

equivalently in both states as a protective factor against having another child.

7 Influence of Socio-Economic Factors on Total Children Ever Born

7.1 Methodology

In this section we shall rigorously analyze and compare the socioeconomic determinants of fertility between two demographically different states in India: Bihar and Kerala. Utilizing unit-level data from the National Family Health Survey (NFHS), the goal is to move beyond simple descriptive statistics to establish how the predictive power of key factors like **Age, Wealth, Educational Attainment, Place of Residence(Rural/Urban)** differ in each context and also provide a comparative analysis between the chosen states. The resulting analysis provides critical insights into the accelerating demographic transition based on the current state of social development.

The objective of this statistical exercise is to model the **total number of children ever born (v201) to a woman** using key socioeconomic and demographic predictors: **Age, Wealth, Educational Attainment, Place of Residence(Rural/Urban)**. We utilize the **Quasi-Poisson regression model** because v201 is count data, and this specific model handles the pervasive **overdispersion** or **underdispersion** (where variance is not equal to the the mean) common in fertility data, ensuring reliable coefficient estimates. Crucially, the model was fitted using the **svyglm** function, which rigorously adjusts for the **complex 2-stage cluster sampling design** of the NFHS. This is accomplished by incorporating the **v001 (IDs)** and **v005 (sample weights)** into the variance calculation. This stabilization of the standard error is vital for generating statistically valid p-values and confidence intervals. Finally, we convert the model's logarithmic coefficients into **Incidence Rate Ratios (IRRs)**, which provide a clear, numerical measure of the multiplicative influence each factor has on the expected number of children.

7.2 Salient Findings and Discussion

To ensure the trustworthiness of our results, we explore the goodness of fit at the very beginning of our analysis. For complex survey GLMs like the Quasi-Poisson, goodness of fit cannot be verified using standard log-likelihood or AIC metrics. Instead, we examine the estimated dispersion parameter ($\hat{\phi}$) and calculate the **Pseudo- R^2** , a measure of proportional variance explained. The Pseudo- R^2 is defined as: $1 - (\text{Residual Deviance}/\text{Null Deviance})$. In social science and demographic studies, R^2 values above 0.30 are considered strong. Our models show excellent explanatory power: the Bihar model achieved an R^2_{Pseudo} of **0.498** and the Kerala model achieved **0.405**. Given these high values, which capture nearly half of the total variance in fertility using just four predictors, we conclude that the models are highly reliable and it is valid to draw strong inferences regarding the socioeconomic drivers in both states.

Pearson Residuals vs. Fitted Values (Model Diagnostic) Checking for systematic patterns in errors (Quasi-Poisson GLM)

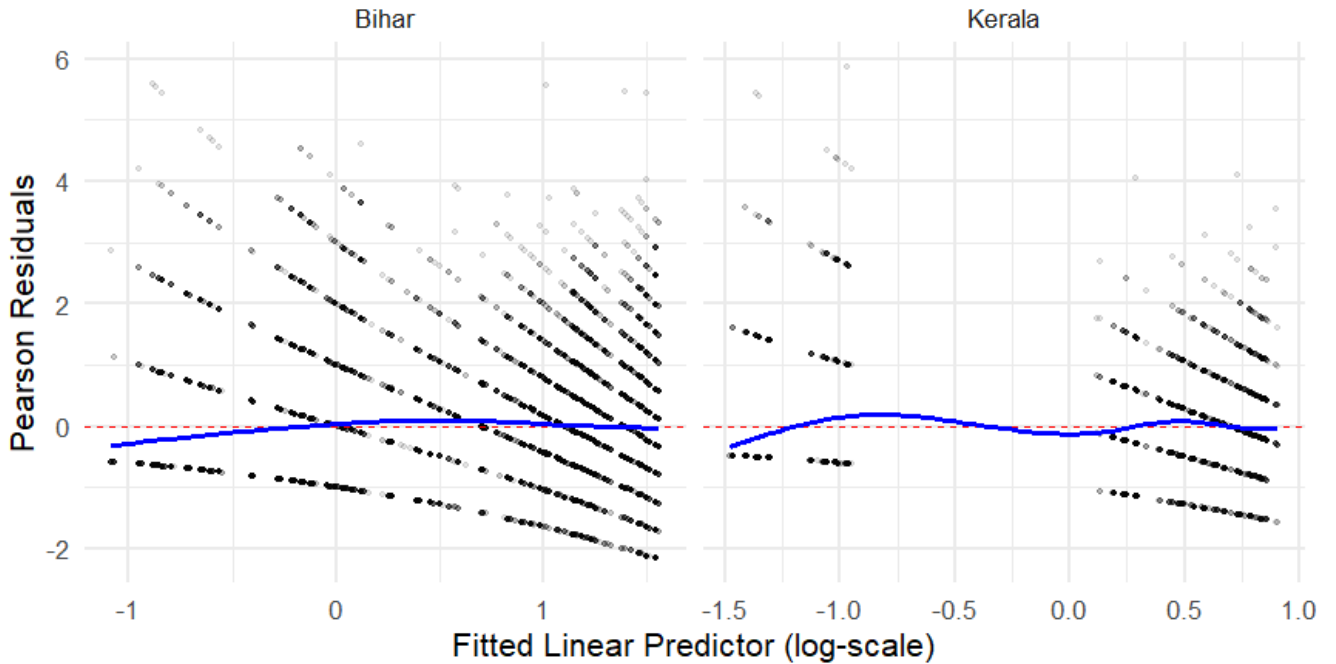


Figure 4: *Pearson Residuals vs. Fitted Values (Model Diagnostic)*

The **Pearson Residuals Plot** (Figure 4) is used to verify the key assumptions of the Generalized

Linear Model (GLM) and validate its fit before drawing major inferences. The plot compares the standardized errors (**Pearson Residuals**) against the model's predicted values (**Fitted Linear Predictor**). The distinct pattern of horizontal bands is expected in **count data** (like Children Ever Born) since the outcomes are limited to discrete integers. The $y = 0$ axis (**Red Dashed Line**) represents a perfect prediction, and the **Blue Loess Line** shows the smoothed average trend of the errors. For a good fit, the Blue Line should be flat and centered on zero. The plot confirms model validity: the blue lines in the both the panels are nearly flat and centered just around the red dotted lines, proving the errors are random and that the model is a good fit.

The fitted models confirm the validity of the Quasi-Poisson choice, as evidenced by the estimated dispersion parameters ($\hat{\phi}$) of 0.825 (Bihar) and 0.553 (Kerala). These estimated dispersion parameters illustrate that the models have successfully accounted for the difference in variance and mean in fertility counts, ensuring the reliability of the calculated standard errors and t -statistics.

The two states exhibit a stark contrast in their statistical predictors: In the regression fit for Bihar, almost every single variable designed to capture socioeconomic status (**all wealth factors** and **all education levels**, save for the reference categories) is highly significant ($p \ll 0.05$). This indicates that, in a high-fertility regime, each marginal step up the socioeconomic ladder yields a statistically measurable reduction in expected children. In contrast, the Kerala Model, all wealth factor levels and all education levels below "Higher" are statistically insignificant ($p > 0.05$). This signifies that basic wealth and education no longer predict differences in family size, as low fertility has become the statistical norm across these strata.

The table below presents the coefficient estimates from the Quasi-Poisson regression models for both states. The significance levels clearly distinguish the active socioeconomic drivers in Bihar from the saturated drivers in Kerala.

The exponentiated intercepts underscore the vast difference in demographic starting points: The

Table 4: Model 1: Bihar Fertility Determinants ($\hat{\phi} = 0.825$)

Predictor	Estimate	t value	P-value
(Intercept)	0.1332	5.765	***
Age 25-34	1.1093	72.042	***
Age 35-49	1.4263	89.883	***
<i>Wealth Factor (Ref: Poorest)</i>			
Poorer	-0.0450	-5.569	***
Middle	-0.0863	-8.187	***
Richer	-0.1526	-9.395	***
Richest	-0.2853	-12.996	***
<i>Education Factor (Ref: No Ed)</i>			
Incomp. Primary	-0.0786	-8.067	***
Incomp. Secondary	-0.2455	-26.618	***
Comp. Secondary	-0.9166	-11.341	***
Higher	-0.6857	-27.627	***
Residence Rural	-0.0142	-0.894	0.371

Signif. codes: *** p<0.001

Table 5: Model 2: Kerala Fertility Determinants ($\hat{\phi} = 0.553$)

Predictor	Estimate	t value	P-value
(Intercept)	-1.0028	-9.398	***
Age 25-34	1.5987	32.179	***
Age 35-49	1.7995	36.322	***
<i>Wealth Factor (Ref: Poorest)</i>			
Poorer	-0.1052	-1.138	0.256
Middle	-0.0407	-0.479	0.632
Richer	0.0095	0.110	0.912
Richest	0.0581	0.671	0.502
<i>Education Factor (Ref: No Ed)</i>			
Incomp. Primary	0.0433	0.710	0.478
Incomp. Secondary	-0.0231	-0.408	0.683
Comp. Secondary	-0.0316	-0.503	0.615
Higher	-0.3691	-6.368	***
Residence Rural	0.0087	0.466	0.642

Signif. codes: *** p<0.001

baseline expected number of children for the youngest, poorest, least-educated woman is ≈ 1.14 in Bihar ($\exp(0.1332)$) but only ≈ 0.37 in Kerala ($\exp(-1.0028)$). This confirms that Kerala has successfully delayed the onset of childbearing across all socioeconomic levels. The insignificance of the Residence Factor (Rural vs. Urban) in both models ($p > 0.37$) suggests that when individual characteristics like wealth and education are properly controlled for, the location's inherent classification (Urban/Rural) does not independently predict fertility—it is the underlying economic and educational resources of the household that truly matter. Tables 4 and 5 have a list of all coefficients and their p-values from which all inferences have been made.

The **Incidence Rate Ratio (IRR)** is the multiplicative change in the expected count of the outcome variable (Children Ever Born) associated with a one-unit change in a predictor. It is calculated by exponentiating the coefficient ($\exp(\beta)$), where an IRR of **1.0** indicates no effect. **Interpretation of Incidence Rate Ratios (IRRs) and Demographic Comparison.** The comparison of IRRs reveals two fundamentally different demographic regimes. In the Bihar model, almost every factor's IRR is highly significant ($p < 0.001$), confirming a steep socioeconomic gradient. For instance, moving to the Richest quintile is associated with a 24.8% decrease

Table 6: Comparative Incidence Rate Ratios (IRR) for Children Ever Born

Predictor (Reference)	Bihar IRR	Change	Sig.	Kerala IRR	Change	Sig.
A. Age Group (vs. 18-24)						
Age Group 25-34	3.03	+203%	***	4.95	+395%	***
Age Group 35-49	4.16	+316%	***	6.05	+505%	***
B. Wealth Factor (vs. Poorest)						
Wealth Factor Poorer	0.96	−4.4%	***	0.90	−10.0%	$p = 0.256$
Wealth Factor Middle	0.92	−8.3%	***	0.96	−4.0%	$p = 0.632$
Wealth Factor Richest	0.75	− 24.8%	***	1.06	+6.0%	$p = 0.502$
C. Education Factor (vs. No Ed)						
Education Factor Incomp. Sec.	0.78	−21.8%	***	0.98	−2.3%	$p = 0.683$
Education Factor Comp. Sec.	0.40	−60.0%	***	0.97	−3.1%	$p = 0.615$
Education Factor Higher	0.50	− 49.6%	***	0.69	− 30.9%	***
Residence Factor Rural	0.99	−1.4%	$p = 0.371$	1.01	+1.0%	$p = 0.642$

Change represents the percentage increase ($IRR - 1$) or decrease ($1 - IRR$) in expected children compared to the reference group.

Sig. codes: *** $p < 0.001$. Variables without stars are statistically insignificant ($p > 0.05$).

in expected children, and obtaining Complete Secondary Education reduces expected children by a staggering 60.0% compared to the baseline. This data shows Bihar is still undergoing a demographic transition that can be immensely catalysed by effective policy implementation focused on wealth and education. Conversely, the Kerala model demonstrates a largely stabilized transition: all IRRs for wealth and lower education levels are clustered near 1.0 (e.g., Richest IRR ≈ 1.06 , $p = 0.502$), meaning these factors are not statistically significant predictors of fertility differences. The only major significant socioeconomic factor is Higher Education, which provides a 30.9% reduction (IRR = 0.69). This evidence demonstrates that Kerala has already achieved a state of demographic saturation, where universal access to low-fertility norms and services has decoupled family size from basic socioeconomic status. The Place of Residence is insignificant in both models, with IRRs hovering around 1.0 (Bihar IRR = 0.99, Kerala IRR = 1.01), indicating that the Urban/Rural classification holds no independent predictive power over fertility once other individual characteristics are controlled. Table 6 has all the IRRs listed for the Kerala and Bihar models for reference.

Figure 5 is the **Confidence Interval Plot** for all Incidence Rate Ratios. This visual representation was required because it provides a quick, robust check of **statistical significance** and **effect magnitude** simultaneously, overcoming the limitations of relying solely on P-values. The

Comparative Incidence Rate Ratios (IRR) for Children Ever Born Analysis of Bihar (High Fertility) vs. Kerala (Low Fertility)

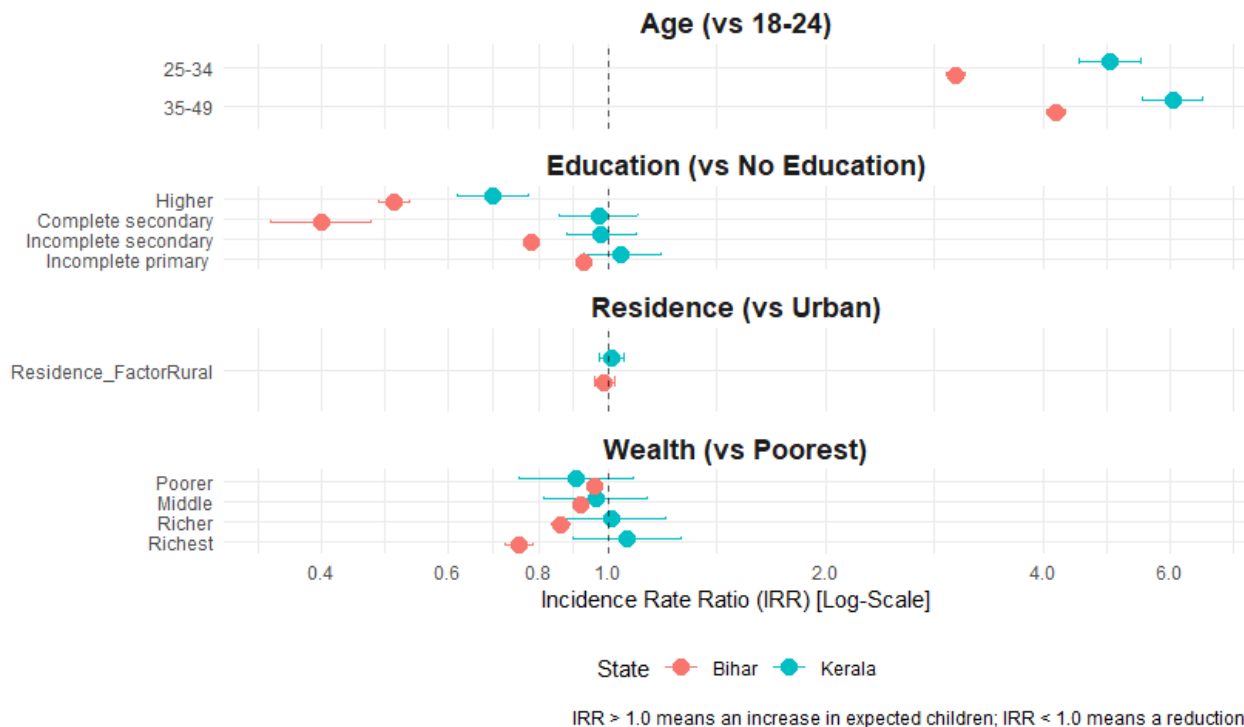


Figure 5: Comparative Incidence Rate Ratios (IRR) with 95% Confidence Intervals

plot shows the IRR (the dot) and the 95% Confidence Interval (the horizontal line); a factor is statistically insignificant if its C.I. **crosses the dashed vertical line at IRR = 1.0** (the "no effect" line). This visualization is perfectly consistent with all our findings: In the Kerala model (Blue), the C.I.s for Wealth and all lower Education levels clearly cross the 1.0 line, visually confirming their insignificance. In contrast, the C.I.s for the corresponding factors in the Bihar model (Red) are tightly grouped and do not cross the 1.0 line, definitively proving that those factors are statistically reliable drivers of fertility reduction in the high-fertility state.

7.3 Conclusion

This comparative analysis, utilizing survey-adjusted Quasi-Poisson regression on NFHS data, rigorously quantifies the divergent socioeconomic-fertility regimes of Bihar and Kerala. The findings demonstrate that fertility in Bihar remains strongly coupled to socioeconomic status, with all wealth and education levels serving as highly significant predictors of fertility reduction (e.g., sec-

ondary education IRR 0.40, $p < 0.001$). This confirms a high-fertility context highly responsive to broad socioeconomic development. Conversely, Kerala’s model demonstrates a saturated demographic transition, where these same factors are statistically insignificant ($p > 0.05$) and fertility is largely decoupled from basic economic status, with only **Higher Education** (IRR 0.69, $p < 0.001$) retaining predictive power. As the rural/urban classifier was insignificant in both states, these results invalidate a uniform national policy, indicating that interventions must be state-specific: focused on fundamental educational and economic upliftment in Bihar, whereas Kerala’s policy context has already transcended these drivers.

The findings of this analysis underscore that India’s fertility transition is far from uniform, necessitating differentiated and context-driven policy responses. In **Bihar**, where fertility remains strongly linked to socioeconomic status, policy must focus on accelerating the demographic transition through social development. Expanding access to **female secondary and higher education**, strengthening **income and employment opportunities** among lower wealth quintiles, and integrating **family planning awareness** within educational and community programs are essential measures. These interventions target the root determinants that statistically drive fertility reduction. In contrast, **Kerala** represents a mature demographic regime where fertility has already stabilized at low levels and is largely decoupled from wealth and basic education. The state’s policies should therefore shift toward managing the consequences of sustained low fertility—addressing **population aging**, ensuring **reproductive autonomy**, enhancing **childcare and health support systems**, and promoting **gender-equitable labor participation**. Overall, these results highlight that a **uniform national fertility policy is ineffective**; instead, India must adopt a **state-specific demographic planning approach** that aligns policy design with each region’s stage in the fertility transition.

8 Determinants of Improved Sanitation in Household

8.1 Methodology

Under this section, we aim to investigate the factors predicting a household's access to **improved sanitation**. For this purpose, we use the household data from the NFHS-5 IR file. The main factors of interest here whose influence we will be testing are the **household wealth index**, the **respondent's education**, and **place of residence**, all of which are categorical variables, and a continuous variable: the **household head's current age**.

To account for the complex survey design, survey design objects were created for Bihar (State 10), Kerala (State 32), and the combined sample, specifying IDs (`v001`) and sampling weights (`v005`). Weighted logistic regression models with a **quasi-binomial** family (to handle potential overdispersion) were fitted separately for Bihar and Kerala to estimate adjusted odds ratios (AORs) for wealth, education, residence, and age.

Subsequently, a combined model was fitted to the pooled data. This model included interaction terms for **wealth-by-state** and **education-by-state**, as well as for all other control variables. This was done to formally test if the effects of these predictors on sanitation access varied significantly between the two states. We used a formal **Wald test** (via the `regTermTest` function) to assess the statistical significance of these interaction terms.

Finally, **predicted probabilities** from the interaction model were generated (using `emmeans` and `ggpredict`) and visualized to illustrate any state-wise differences in the effects of these key predictors.

8.2 Salient Findings and Discussion

Our investigation into the determinants of improved sanitation access reveals striking differences between Bihar and Kerala. A crude analysis of the weighted prevalence of having an improved

household toilet provides an initial look at this divergence. As shown in Table 7, the prevalence of improved sanitation in Bihar for the poorest quintile starts at 43.0% and shows a steep monotonic increase to 66.7% for those with higher education.

In contrast, Kerala’s crude prevalence is substantially higher and shows almost universal access, starting at 97.5% for those with no education and remaining high for all other groups. The standard error for Kerala’s “No Education” quintile is notably large (0.0150) compared to Bihar’s (0.0077), indicating significant variance for this subgroup in Kerala.

Table 7: *Weighted Prevalence of Improved Toilet Access by Education*

Woman’s Education	Bihar (State 10) Prevalence (%)	Bihar (SE)	Kerala (State 32) Prevalence (%)	Kerala (SE)
No Education	43.0%	0.0077	97.5%	0.0150
Primary	52.8%	0.0112	98.9%	0.0059
Higher	66.7%	0.0072	98.3%	0.0029

To understand these patterns beyond this crude analysis, we fitted separate weighted logistic regression models for each state, controlling for wealth, education, residence, and age. The Adjusted Odds Ratios (AORs) in Table 8 reveal that the drivers of sanitation access appear to be fundamentally different in the two states.

Table 8: *Adjusted Odds Ratios (AORs) for Improved Sanitation Access (from Models 1 & 2)*

Factor	AOR for Bihar (p-value)	AOR for Kerala (p-value)
Wealth (Ref: Poorest)		
Poorer	4.43 ($p < .001$)	4.57 ($p = .004$)
Middle	13.52 ($p < .001$)	7.51 ($p < .001$)
Richer	39.79 ($p < .001$)	17.54 ($p < .001$)
Richest	60.77 ($p < .001$)	18.79 ($p < .001$)
Education (Ref: No Ed.)		
Primary	1.12 ($p = 0.016$)	1.40 ($p = 0.630$)
Higher	1.29 ($p < .001$)	0.57 ($p = 0.403$)
Residence (Ref: Urban)		
Rural	0.70 ($p < .001$)	0.91 ($p = 0.832$)
HH Head Age (continuous)	1.00 ($p = 0.336$)	1.00 ($p = 0.961$)

Upon analyzing the above data, we observe that in **Bihar**, household **wealth**, a woman’s **education**, and **residence** are all highly significant predictors. In contrast, for **Kerala**, **wealth** remains a very strong, significant predictor, but the effects of **education** and **residence** are not

statistically significant. **Age** of the household head is not a significant factor in either state.

Given these stark apparent differences in the stratified models, we want to formally test whether the effects of wealth and education vary across states. To this end, we used a combined interaction model on data from both Bihar and Kerala. The formal Wald tests for these interactions tell a more nuanced story:

- The test for the **state-by-wealth** interaction yields an F-statistic of 2.33, with $p = 0.054$.
- The test for the **state-by-education** interaction yields an F-statistic of 2.35, with $p = 0.095$.

At a conventional $\alpha = 0.05$ significance level, we fail to reject the null hypothesis for both interactions. This indicates that while the AORs in Table 8 *appear* different, we do not have sufficient statistical evidence to conclude that the effects of wealth or education on sanitation access are truly different between Bihar and Kerala. The visible differences are statistically indistinguishable from what might be observed due to random chance, a finding underscored by the borderline p-value for the wealth interaction ($p = 0.054$).

This leads us to the plots from the interaction model, which must be interpreted with this new context.

The interaction plot for wealth (Figure 6) visually suggests a difference, particularly in the lower quintiles. However, as our formal Wald test ($p = 0.054$) showed, this visual gap is not statistically significant. The wide confidence intervals for Kerala’s poorer quintiles, which overlap with Bihar’s, reflect this statistical uncertainty. The primary takeaway is that for both states, wealth is a powerful positive predictor, and the statistical evidence is not strong enough to claim the "return on wealth" is different between the states.

Similarly, the education plot (Figure 7) shows a large visual gap between the two states. This aligns with our Wald test ($p = 0.095$), which found no significant interaction. This suggests that the large gap is a main effect of state (as seen in the interaction model, stateKerala p-value is 0.007), not a differential effect of education. The plot also confirms the findings from our stratified models: in Bihar, the move from “No Education” to “Primary” and “Higher” shows a small, significant increase (red dots), while in Kerala, the line is flat, showing no effect (blue dots).

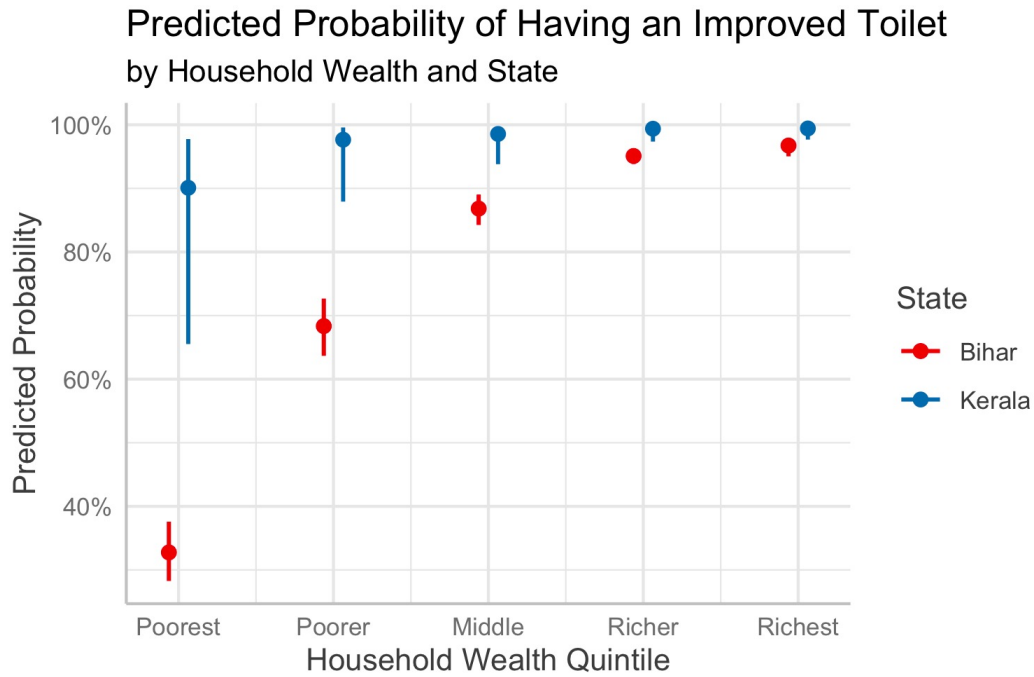


Figure 6: Predicted Probability of Having an Improved Toilet by Household Wealth and State.

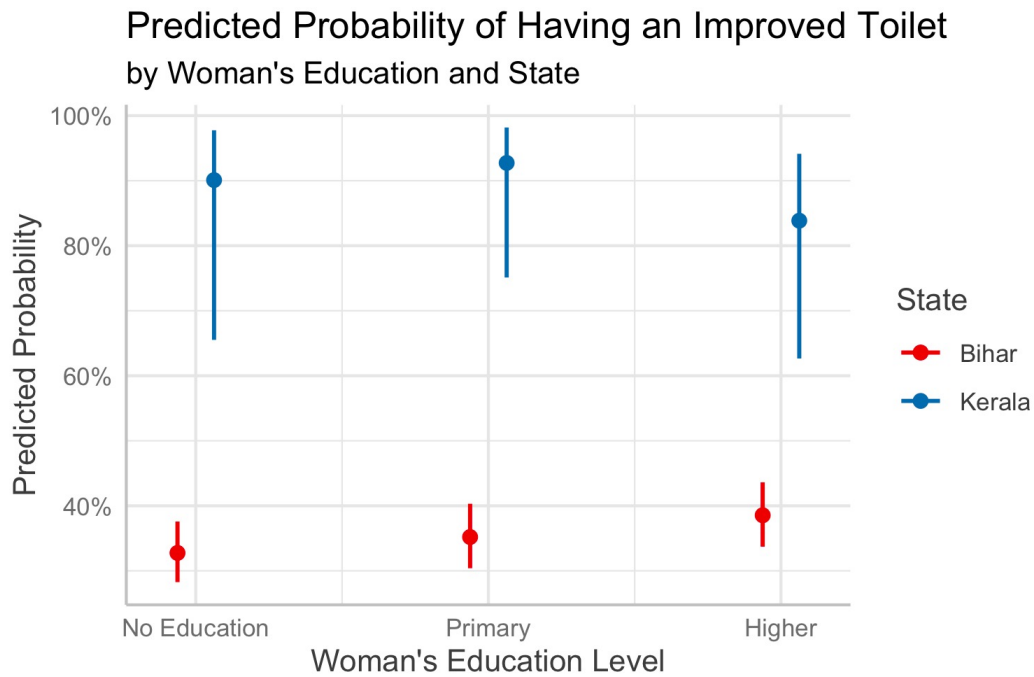


Figure 7: Predicted Probability of Having an Improved Toilet by Woman's Education and State.

8.3 Conclusion

This analysis demonstrates the critical impact of a “ceiling effect” when analyzing high-prevalence outcomes, such as Kerala’s near-universal sanitation. While a strong, clear wealth gradient is evident in Bihar (e.g., Richest AOR 60.77 vs. Poorest, $p < .001$), the effect of education is small but significant (Higher AOR 1.29, $p < .001$). In contrast, Kerala’s saturation regime neutralizes these factors: the wealth effect is diminished (Richest AOR 18.79, $p < .001$), and the education effect becomes statistically insignificant ($p > 0.05$).

A formal interaction model, designed to test this apparent divergence, proves critical. It reveals that while the visual difference in development pathways is large, the statistical evidence for a true interaction is insufficient for both predictors. The test for the between state wealth interaction is borderline ($F=2.33$, $p=0.054$), and the test for the between state education interaction is not significant ($F=2.35$, $p=0.095$). Therefore, we fail to reject the null hypothesis for either.

The analysis confirms that while wealth is a powerful determinant in Bihar and education is a minor one, the influence of both is substantially diminished in Kerala’s high-access context. However, it also demonstrates that the underlying statistical relationships (the mechanisms) for these factors cannot be proven to be statistically different between the states. This suggests Bihar is on a steep developmental curve where wealth is paramount, while Kerala has largely transcended these basic drivers.

9 Analysis of the Relationship between Wealth Quintile and Education Level

9.1 Methodology

This study examines the effect of household wealth on attaining *Secondary or More* education in Bihar (high fertility, low education) and Kerala (low fertility, high education) using NFHS-5 unit-level data. Education (v149) is recoded into *Primary or Less* (coded as 0-2) vs. *Secondary or More* (coded as 3-5), and wealth (v190) into three groups: *Poorer* (coded as 1-2), *Middle* (coded as 3), and *Richer* (coded as 4-5). All analyses incorporate NFHS sampling weights, stratification,

and clustering to ensure population-representative inference.

A key methodological issue is Kerala's near-universal schooling rates, which create a **ceiling effect** that renders conventional logistic regression unstable and leads to inflated odds ratios.

Our empirical strategy proceeds in four distinct steps. First, a **Descriptive Analysis** is conducted by estimating and plotting the weighted prevalence of secondary-or-higher education by wealth group, which serves to visually highlight the **ceiling effect** of near-universal attainment in Kerala. Second, we fit a standard survey-adjusted **Logistic Regression** model to demonstrate how the resulting **odds ratios (ORs)** become distorted and substantively misleading under such high-prevalence conditions. Third, as an **Interaction Test Demonstration**, we fit a pooled logistic model including a wealth-by-state interaction term and apply a formal **Wald test**. This step shows the test's failure to detect a true interaction, which we attribute to the logistic model's underlying misspecification. Finally, we implement a **Modified Poisson Regression** using survey-adjusted Poisson models with robust standard errors to correctly estimate **Relative Risks (RRs)**, which provide a stable and interpretable measure of effect that remains valid even when the outcome is common. **Relative Risk (RR)** is defined as the ratio of the probability of achieving "Secondary or More" education for a specific wealth group ('Middle' or 'Richer') to the probability of the same outcome for the baseline 'Poorer' group. This ratio is directly obtained by exponentiating the coefficients (β) from the **quasipoisson** log-linear model, such that $RR = e^\beta$.

This framework shows that while logistic regression exaggerates wealth gradients in Kerala, the modified Poisson model yields valid and consistent estimates across both states.

9.2 Salient Findings and Discussion

The results highlight strong cross-state differences in both educational patterns and model performance. Bihar displays a clear wealth gradient in educational attainment, while Kerala shows near-universal schooling, creating a **ceiling effect** that challenges conventional logistic regression. Figure 8 shows the weighted proportion of individuals with Secondary-or-More education by wealth group. Bihar Attainment rises sharply with wealth (from around 40% among the poorer to > 85% among the richer), reflecting a strong gradient. Whereas Kerala Education levels are already high

(around 82% in the poorer group) and approach universality (around 98% among the richer). This saturation results in a **ceiling effect**, limiting observed variation.

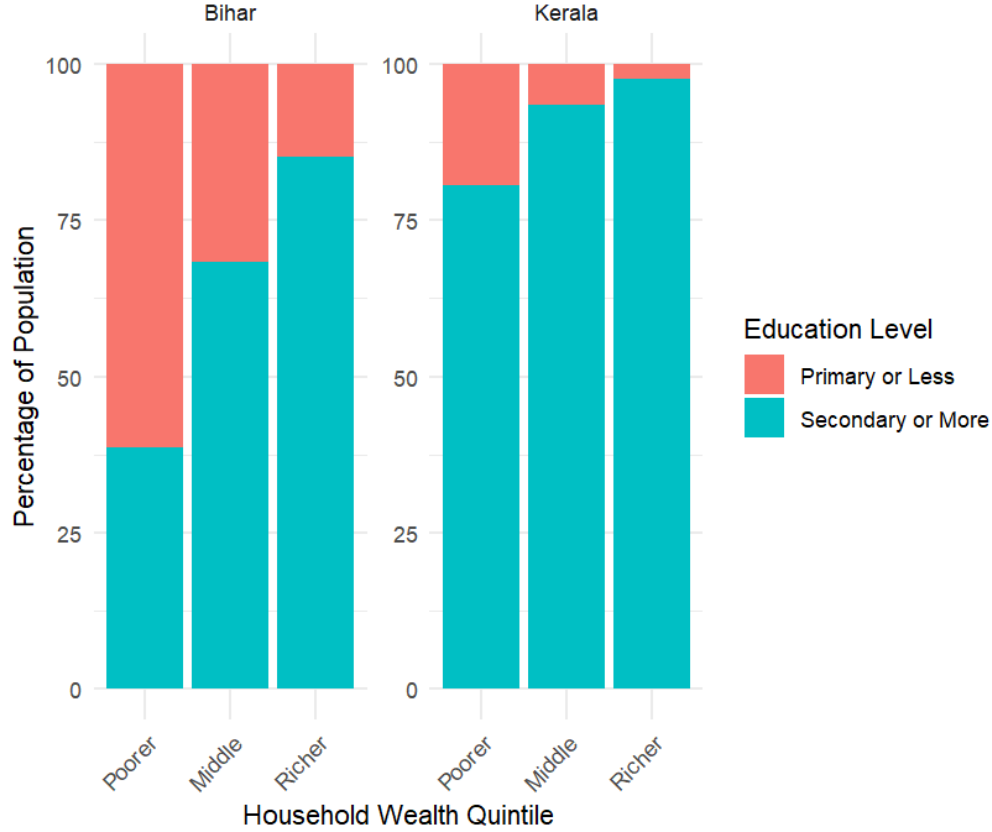


Figure 8: Distribution of Education Level by Wealth and State

A standard survey-weighted logistic regression model, run separately for each state, produced Odds Ratios (ORs) that directly contradicted the visual evidence. The model's failure is proven by comparing the model's reported Odds Ratios to the "**real-world**" Odds Ratios calculated directly from the descriptive data 8.

Table 9: Comparison of Real-World Odds Ratios vs. Faulty Model-Reported Odds Ratios

State	Descriptive Probabilities (Richer vs. Poorer)	Real-World OR (Calculated from Probs.)	Model-Reported OR (From <code>svyglm</code>)
Bihar	≈ 86% vs. ≈ 39%	$\frac{0.86/(1-0.86)}{0.39/(1-0.39)} \approx 9.6$	9.1 (A near-perfect fit)
Kerala	≈ 99.5% vs. ≈ 81%	$\frac{0.995/(1-0.995)}{0.81/(1-0.81)} \approx 46.7$	10.2 (A total model failure)

As shown in Table 9, the model for Bihar is an excellent fit (9.6 vs. 9.1). The model for Kerala,

however, fails completely. It cannot compute the true, massive Odds Ratio of 46.7 and instead produces a nonsensical, unstable estimate of 10.2.

Although mathematically correct, these ORs are **substantively misleading**: in Kerala, the true increase is only about 16 percentage points (82% to 98%), but the odds transformation magnifies this into a large ratio. Consequently, the logistic model incorrectly implies a similarly strong wealth effect in both states.

A wealth–state interaction test further fails (p -value=0.69), not because effects are equal, but because the model is unsuitable under extreme outcome prevalence.

The modified Poisson model (quasipoisson family) generates **Relative Risks (RRs)**, which remain valid under high prevalence (Table 11).

Table 10: Robust Poisson Model Coefficients (Reference Group: 'Poorer')

Model 1: Bihar				Model 2: Kerala			
Predictor	Estimate	t value	P-value	Predictor	Estimate	t value	P-value
(Intercept)	-0.9499	-84.36	***	(Intercept)	-0.2156	-10.24	***
Middle	0.5697	38.95	***	Middle	0.1482	7.17	***
Richer	0.7891	56.88	***	Richer	0.1924	9.14	***

Signif. codes: *** $p < 0.001$

Table 11: Relative Risks (RRs) from Modified Poisson Model

State	Term	RR
Bihar	Middle vs. Poorer	1.77
	Richer vs. Poorer	2.20
Kerala	Middle vs. Poorer	1.16
	Richer vs. Poorer	1.21

In Bihar: A person from the Middle wealth group is 1.77 times as likely (or 77% more likely) to have a secondary education or more, compared to a person from the Poorer group. A person from the Richer wealth group is 2.20 times as likely (or 120% more likely) to have that same level of education, compared to the Poorer group. So, Money is a major factor. There is a clear and strong wealth gradient, where your chances of being highly educated increase dramatically as your household wealth goes up.

But in Kerala: A person from the Middle wealth group is only 1.16 times as likely (or 16% more likely) to have a secondary education or more, compared to the Poorer group. A person from the

Richer wealth group is only 1.21 times as likely (or 21% more likely) to have that same education, compared to the Poorer group.

The `quasipoisson` model estimates a dispersion parameter, ϕ , to correct for violations of the standard Poisson assumption ($\phi = 1$) where the mean equals the variance. The results indicate significant underdispersion ($\phi < 1$) for both states: approximately 0.49 for Bihar and 0.04 for Kerala.

This underdispersion is expected, as the model is applied to a binary 0/1 outcome. For such data, the mean is the probability p , while the variance is $p(1 - p)$. Since $p(1 - p) < p$, the variance is inherently less than the mean. The model correctly estimates $\phi \approx (1 - p)$: the severe underdispersion in Kerala (0.04) reflects its high “ceiling effect” (i.e., $p \approx 0.96$), while Bihar’s moderate underdispersion (0.49) reflects its probability $p \approx 0.51$. This estimation correctly scales the standard errors, validating the relative risk calculations.

These RRs align with observed data: wealth strongly predicts educational attainment in Bihar but has only a modest association in Kerala. Unlike ORs, RRs accurately reflect the true substantive difference between states. Logistic regression dramatically overstates Kerala’s wealth effect due to the ceiling effect. The modified Poisson model provides correct, interpretable, and policy-relevant estimates, confirming a large wealth gradient in Bihar and a negligible one in Kerala.

9.3 Conclusion

This analysis demonstrates the critical impact of model choice when analyzing high-prevalence outcomes, such as Kerala’s near-universal education. While a strong, clear wealth gradient in educational attainment is evident in Bihar (RR: 2.20 for Richer vs. Poorer), this effect is minimal and substantively negligible in Kerala (RR: 1.21), where a ceiling effect dominates. Standard logistic regression proves unsuitable in this context, failing to detect the true interaction and producing distorted Odds Ratios that misleadingly inflate Kerala’s wealth effect. The modified Poisson model, however, successfully provides stable and interpretable Relative Risks that align with the descriptive data, confirming that while wealth is a powerful determinant of education in Bihar, its influence has been largely neutralized in Kerala.

10 The Impact of Household Wealth on the Likelihood of Early Childbearing

10.1 Methodology

We aim to investigate the influence of household wealth on the likelihood of early childbearing (among women aged 15-21). We define early childbearing as the occurrence of at least one live birth among adolescent women aged 15-21. To ensure comparability, the dataset was restricted to respondents aged 15-21 and variables were recoded appropriately. The main factor whose influence we wanted to test was the **household wealth index**, which was treated as a categorical variable ('1': poorest, '2': poor, '3': middle, '4': rich, '5': richest). Along with this, some additional control variables as **education** and **place of residence** (rural/urban) was used. Due to insufficient data available, we were forced to recode the education variable as a binary variable - coded 0 for any education lower than "complete secondary" and 1 for "complete secondary or higher". Place of residence was coded as '1': Urban and '2': Rural.

We used survey design objects for Bihar, Kerala, and the combined sample, to account for the complex survey design. Further, weighted logistic regression models with a binomial family were fitted separately for Bihar and Kerala to estimate the adjusted odds ratios (AORs) for household wealth, education, and residence. A combined model including a wealth-state interaction term was also fitted to test for the varying effects of wealth across different states. Finally model results were summarized through AORs and predicted probabilities were visualized through a plot to illustrate statewise differences in the wealth-fertility relationship.

10.2 Salient Findings and Discussion

If we just find the prevalence rate of early childbearing in both the states, we wind up with the following table. Table 12 apparently tells us that Kerala has a much lower overall prevalence of early childbearing than Bihar. However, it reveals less or no information about the influence of various factors that contribute to this difference.

Table 12: *Prevalence of Early Childbearing (Ages 15-21) by Wealth Quintile*

Wealth Quintile	Bihar (State 10) Prevalence (%)	Kerala (State 32) Prevalence (%)
1 (Poorest)	21.0%	9.7%
2 (Poorer)	17.7%	2.8%
3 (Middle)	15.3%	6.4%
4 (Richer)	11.9%	5.8%
5 (Richest)	8.3%	4.8%

Upon calculating individual Adjusted Odds Ratios for early childbearing using weighted logistic regression (to adjust for survey design) with a binomial family, we get the following table 13. Here, we have taken the base line as wealth code 1 (poorest), education code 0 (education lower than completion of secondary school), and place of residence as urban.

Table 13: *Adjusted Odds Ratios for Bihar and Kerala*

Factor	AOR for Bihar	AOR for Kerala
wealth2	0.827	0.289
wealth3	0.739	0.752
wealth4	0.579	0.730
wealth5	0.426	0.633
education	0.574	0.673
residence2	1.073	1.468

This tells us that **wealth** and **education** are excellent protective factors against early childbearing. However, although early childbearing shows a marked decrease with increase in the household wealth index, the same cannot be directly said for Kerala. One can note that the effect in Kerala is not a smooth, monotonic decrease. Therefore, to determine whether the effect of wealth varies between the two states, we conducted a combined regression analysis using data from both Bihar and Kerala by considering the "Bihar, poorest, uneducated, urban women" category as reference.

Table 14 presents a detailed collection of the findings. We observe here that the interaction effects between the household wealth index and state is statistically insignificant here, reflecting the near same effect of wealth in both the states to reduce early childbearing. Further, we see that educa-

tion too plays a very important role, and hence must not be ignored.

The coefficient for Kerala is negative but not statistically significant ($\beta = -0.89, p = 0.40$), indicating that after adjusting for wealth, education, and place of residence, the difference in early childbearing between Bihar and Kerala is no longer significant. This implies that Kerala's lower crude prevalence of early childbearing is largely explained by its more favourable socioeconomic composition rather than by state-level effects. Once individual-level factors are controlled for, state-level differences in early fertility behaviour become statistically indistinguishable.

Table 14: *Combined Regression Analysis on data from both Bihar and Kerala*

Variable	Estimate (Log-Odds)	p-value
(Intercept)	-1.40608	< 2e-16
wealth2	-0.18891	0.002987
wealth3	-0.29837	0.000269
wealth4	-0.53619	4.90e-06
wealth5	-0.83492	0.000267
state32	-0.88976	0.402035
education	-0.52955	3.22e-09
residence2	0.11215	0.260580
wealth2:state32	-0.98157	0.413980
wealth3:state32	0.05678	0.958506
wealth4:state32	0.21806	0.840459
wealth5:state32	0.32544	0.767665

We also performed a formal statistical Wald test to check for an interaction effect between the household wealth index and state. The test yielded an F -statistic of 1.35 (on 4 and 2251 df), corresponding to a p -value of 0.248. Since this p -value is greater than our 0.1 significance level, we fail to reject the null hypothesis. We, therefore, conclude that there is no statistically significant interaction between wealth and state affecting the likelihood of early childbearing.

Further, everything we have described so far can be summarized using the following plot of the predicted probability of early childbearing by wealth and state. The plot visually confirms the F -test. While the patterns look different—Bihar shows a smooth gradient while Kerala shows a 'poverty cliff' (a steep drop from quintile 1 to 2)—the wide, overlapping confidence intervals show that these visual differences are not statistically significant. The large confidence interval in

wealth1 variable in Kerala can be explained by the fact that there are less sample points of the above sort for us to analyze in details and hence we end up with a not so good overall estimate for the wealth1 variable in case of Kerala.

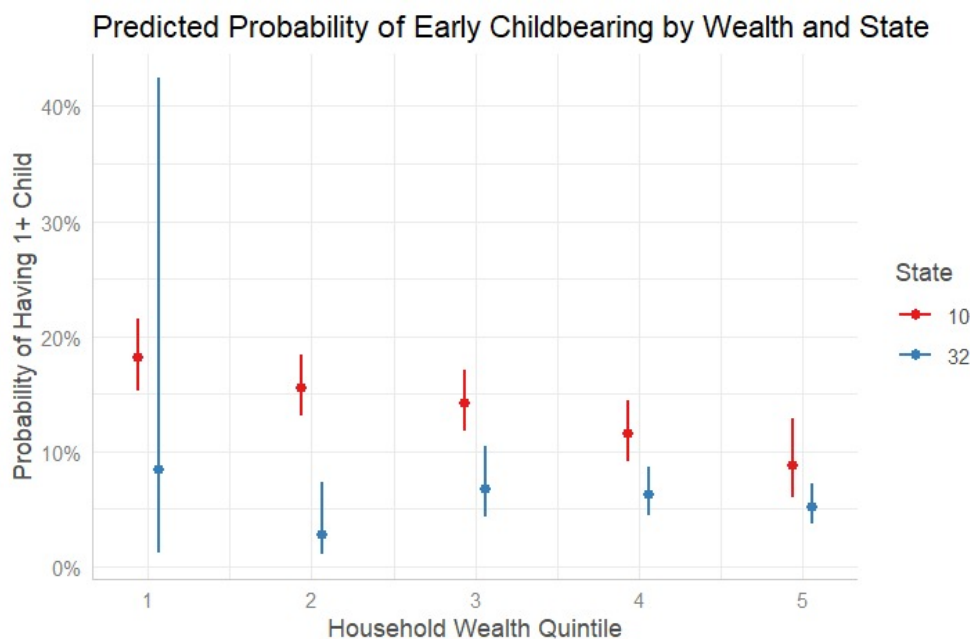


Figure 9: Predicted Probability of Early Childbearing by Wealth and State

The state-level regressions in 15 reveal two distinct fertility–wealth relationships. In Bihar, household wealth emerges as a strong and consistent predictor of early childbearing — the odds decrease steadily and significantly as we move from the poorest to the richest quintile. Education also exerts an independent protective effect, while urban–rural residence has little influence. In contrast, in Kerala, none of the wealth quintiles are statistically significant, and even though education and residence show weak effects, the overall variation in early childbearing across socioeconomic groups is minimal. This pattern suggests that Kerala, having achieved a high level of social development and near-universal access to education and health services, has effectively delinked fertility behavior from economic status. In such a context, awareness about contraception and family planning is widespread across all wealth strata, resulting in uniformly low rates of early childbearing. Bihar, on the other hand, still reflects a classic socioeconomic gradient, where economic and educational advantages translate into delayed fertility and better reproductive health outcomes.

Table 15: *Result of State Wise Analysis*

Variable	Bihar			Kerala		
	Estimate	Std. Error	p-value	Estimate	Std. Error	p-value
Intercept	-1.365***	(0.113)	< 0.001	-2.510*	(1.114)	0.0247
Wealth (Q2)	-0.190**	(0.064)	0.0029	-1.240	(1.217)	0.3089
Wealth (Q3)	-0.302***	(0.082)	0.0002	-0.285	(1.109)	0.7971
Wealth (Q4)	-0.546***	(0.118)	< 0.001	-0.314	(1.104)	0.7758
Wealth (Q5)	-0.852***	(0.231)	< 0.001	-0.458	(1.113)	0.6810
Education (Higher)	-0.555***	(0.098)	< 0.001	-0.397 [†]	(0.226)	0.0795
Residence (Urban)	0.070	(0.107)	0.5110	0.384 [†]	(0.231)	0.0967

10.3 Conclusion

Our analysis confirms that both higher **household wealth** and **education** are powerful, statistically significant protective factors against early childbearing. While state-specific models suggested a different wealth-fertility relationship—a smooth gradient in Bihar versus a non-monotonic pattern in Kerala—our formal interaction model proved otherwise. The **Wald test** for the wealth-by-state interaction was **not significant** ($p = 0.248$), indicating we cannot conclude the effect of wealth is statistically different between the two states. The state wise analysis however revealed that while the effect of Wealth is Significant in Bihar, it is not so for Kerala, suggesting that Kerala has transitioned state that has detached the economic status effect from fertility. Education, just like Wealth is a lot more significant in Bihar than in Kerala, where it is significant at a significance level of 0.1 but not 0.05. The observed visual differences in the interaction effect, which we found to not be significant are likely attributable to random sampling variation and significant data sparsity in Kerala’s lower wealth quintiles, as evidenced by the plot’s wide, overlapping confidence intervals.

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Table 16: *Distribution of Work by Research Question*

Research Question / Task	Contributor(s)
1. Parity Progression (1-to-2+ Children) Analysis of factors influencing the transition to a second birth.	Rwitobroto Dey (BS2341)
2. Total Children Ever Born Quasi-Poisson analysis of socio-economic determinants of fertility.	Pratyush Patra (BS2336)
3. Improved Sanitation Access Logistic regression on determinants of household sanitation.	Arnab Palit (BS2315)
4. Wealth-Education Relationship Modified Poisson analysis of the wealth-education link.	Aryanil Kundu (BS2318)
5. Early Childbearing Analysis of factors impacting adolescent (age 15-21) childbearing.	Sourodeep Kanjilal (BS2351) Rwitobroto Dey (BS2341)
Project Finalization Data management, report compilation, and L ^A T _E X typesetting.	All Members